A Non-Suspension Polynomial Solution in Gao's Five-Dimensional Middle Branch

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Abstract

For the mixed direction field
$$\Delta = (1, w_1, w_1^2, w_2)^\mathsf{T},$$
Gao's five-dimensional middle branch is governed by the polynomial identity $4cD_0 = D_1^2$ and was conjectured to be rigid up to suspension of a four-dimensional sweep. We give explicit polynomial data satisfying the middle-branch equation and prove that the associated unpadded sweep is not polynomially equivalent to any triangular one-coordinate suspension of a four-dimensional Gao sweep. No determinant normalization is imposed on the suspension: polynomial equivalence to our sweep would force its Jacobian determinant to be a nonzero constant multiple of the square of the Gao sweep parameter, after which Gao's general sweep formula and unique factorization force the two possible base-sweep determinant orders. The proof uses the critical-value closure. For our sweep it is the irreducible threefold
$$(z_1z_3 - z_2^2)^3 z_4^2 - \lambda z_1^6 = 0, \qquad \lambda = \frac{41-38i}{576} \neq 0,$$
which is not ambient-polynomially equivalent to a cylinder. We formulate the theorem independently of the undefined phrase "equivalent to a suspension" in Gao's Open Problem 4.8. Under the source-target polynomial-equivalence interpretation and the triangular one-coordinate suspension class defined here, Open Problem 4.8 therefore has a negative answer.

1 Context and statement of the result

Gao studies polynomial tangent sweeps associated with the mixed direction field
$$\Delta = (1, w_1, w_1^2, w_2)^\mathsf{T}$$
in dimension five. With potentials $\Gamma = (G_2, G_3, G_4)$ and
$$H := G_3 - 2w_1G_2,$$
Gao writes the determinant pencil as
$$\det\bigl(J(\Gamma) + \mu\widetilde{P}\bigr) = D_0 + \mu D_1 + \mu^2 D_2,$$
where
$$\begin{aligned}
D_2 &= -\partial_3 H,\\
D_1 &= (\partial_2 H)(\partial_3 G_4) - (\partial_3 H)(\partial_2 G_4) - \bigl((\partial_1 G_2)(\partial_3 G_3) - (\partial_1 G_3)(\partial_3 G_2)\bigr),\\
D_0 &= \det J(G_2, G_3, G_4).
\end{aligned}$$

The middle branch is normalized by

$$D_2 = c \in \mathbb{C}^\times, \qquad X_1 = -\frac{D_1}{2c}, \qquad 4cD_0 = D_1^2. \tag{1}$$

This is Gao's equation (12) in arXiv:2608.00222v1 [1]. Gao's Theorem 4.6 proves that on the characteristic-invariance locus $\mathcal{W}G_2 = 0$ every solution collapses to a triangular suspension of a four-dimensional bottom-branch sweep, while Remark 4.7 records another explicit graph-coordinate collapse. Open Problem 4.8 asks whether every polynomial solution of (1) is equivalent to a suspension of a four-dimensional sweep [1].

Throughout, "five-dimensional middle branch" follows Gao's padded construction $F_0 : \mathbb{A}^5 \to \mathbb{A}^5$. The critical-value geometry below concerns its unpadded sweep $S : \mathbb{A}^4 \to \mathbb{A}^4$; padding contributes one extra coordinate and one extra factor of γ to the Jacobian determinant. Because the phrase "equivalent to a suspension" is not formally defined in the statement of Open Problem 4.8, we first state a precise theorem that does not depend on that terminology.

Definition 1 (Source-target polynomial equivalence)**.** Two polynomial maps $S_1, S_2 : \mathbb{A}^4 \to \mathbb{A}^4$ are *polynomially equivalent* if

$$S_2 = \Psi \circ S_1 \circ \Phi$$

for polynomial automorphisms $\Phi, \Psi \in \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^4)$.

Definition 2 (Triangular one-coordinate suspension)**.** Let $u = (\gamma_T, a, b)$ and let v be a fourth source coordinate. A *triangular one-coordinate suspension of a four-dimensional Gao sweep* is a polynomial map

$$\Sigma(u, v) = \big(T(u), Q(u, v)\big) : \mathbb{A}^4 \to \mathbb{A}^4,$$

where $T : \mathbb{A}^3 \to \mathbb{A}^3$ is the unpadded sweep associated with an arbitrary four-dimensional Gao tangent-sweep datum and T is independent of v. No Jacobian-determinant normalization and no determinant branch for T are imposed in the definition.

Theorem 3 (Main theorem)**.** *There exists an exact polynomial solution of Gao's five-dimensional middle-branch equation* (1) *whose associated unpadded sweep* $S : \mathbb{A}^4 \to \mathbb{A}^4$ *is not polynomially equivalent to any triangular one-coordinate suspension of a four-dimensional Gao sweep.*

Corollary 4 (Relation to Gao's Open Problem 4.8)**.** *If "equivalent to a suspension of a four-dimensional sweep" in Gao's Open Problem 4.8 is interpreted as source-target polynomial equivalence to the triangular one-coordinate suspension class above, then Open Problem 4.8 has a negative answer.*

Remark 5 (Scope of the theorem)**.** Theorem 3 excludes exactly the class defined above: triangular one-coordinate suspensions whose base is an arbitrary four-dimensional Gao sweep, under source-target polynomial equivalence. It does not claim to classify every object that could informally be called a "suspension." Corollary 4 is therefore intentionally conditional on interpreting Gao's terminology by this precise class.

2 The exact middle-branch witness

Normalize $c = 1$ and $\varphi = 0$, and write

$$x = w_1, \qquad y = w_2, \qquad t = w_3.$$

Set

$$\boxed{G_2 = yt^2, \qquad G_3 = 2xyt^2 - t, \qquad G_4 = \frac{4i}{3}y^3t^3.} \tag{2}$$

Then

$$H = G_3 - 2xG_2 = -t.$$

This witness lies outside Gao's characteristic-invariance collapse locus. Indeed, for $c = 1$ and $\varphi = 0$ one has $\mathcal{W} = \partial_y$, while

$$\mathcal{W}G_2 = \partial_y(yt^2) = t^2 \neq 0.$$

Thus Theorem 4.6 does not apply to the witness.

Proposition 6. *The data* (2) *satisfy Gao's middle-branch conditions with* $c = 1$*:*

$$D_2 = 1, \qquad D_1 = 4(1+i)y^2t^3, \qquad D_0 = 8iy^4t^6,$$

and hence $4D_0 = D_1^2$ *identically.*

Proof. From $H = -t$ one has $D_2 = -\partial_t H = 1$. Since

$$\partial_y H = 0, \qquad \partial_t H = -1,$$

Gao's displayed formula for D_1 reduces to

$$D_1 = \partial_y G_4 - \big((\partial_x G_2)(\partial_t G_3) - (\partial_x G_3)(\partial_t G_2)\big).$$

Now $\partial_x G_2 = 0$, $\partial_x G_3 = 2yt^2$, $\partial_t G_2 = 2yt$, and

$$\partial_y G_4 = 4iy^2t^3.$$

Therefore

$$D_1 = 4iy^2t^3 + 4y^2t^3 = 4(1+i)y^2t^3.$$

A direct 3×3 determinant gives

$$D_0 = \det J(G_2, G_3, G_4) = 8iy^4t^6.$$

Consequently

$$D_1^2 = 16(1+i)^2y^4t^6 = 32iy^4t^6 = 4D_0.$$

□

The middle-branch reconstruction gives

$$X_1 = -\frac{D_1}{2} = -2(1+i)y^2t^3$$

and

$$\begin{aligned}
X_2 &= G_2 + xX_1 = yt^2 - 2(1+i)xy^2t^3, \\
X_3 &= G_3 + x^2X_1 = 2xyt^2 - t - 2(1+i)x^2y^2t^3, \\
X_4 &= G_4 + yX_1 = \left(-2 - \frac{2i}{3}\right)y^3t^3.
\end{aligned} \tag{3}$$

For

$$\Delta = (1, x, x^2, y)^{\mathsf{T}}$$

define the unpadded sweep

$$S(\gamma, x, y, t) = X(x, y, t) + \gamma\Delta(x, y).$$

Proposition 7. *The sweep satisfies*

$$\det JS = \gamma^2.$$

Moreover X has generic rank three; for example

$$\det \frac{\partial(X_1, X_2, X_3)}{\partial(x, y, t)} = (-4 + 12i)y^3 t^6.$$

Thus $\mathrm{Crit}(S) = \{\gamma = 0\}$ *and the critical-value closure has dimension three.*

Proof. Substitution of (3) into the 4×4 Jacobian gives the polynomial identity $\det JS = \gamma^2$. The displayed 3×3 minor is nonzero, hence $\operatorname{rank} JX = 3$ on a dense open subset. Since $\det JS = \gamma^2$, the set-theoretic critical locus is exactly $\{\gamma = 0\}$; scheme-theoretically the determinant vanishes there with multiplicity two. Its image is $X(\mathbb{A}^3)$, whose closure has dimension three by generic rank three. □

3 Elimination and the critical-value threefold

Let (z_1, z_2, z_3, z_4) be target coordinates and set

$$R = z_1 z_3 - z_2^2.$$

Write

$$\kappa = 2(1+i), \qquad \rho = 1 + 2i, \qquad d = -2 - \frac{2i}{3}.$$

From (3),

$$z_1 = -\kappa y^2 t^3, \qquad z_4 = d y^3 t^3.$$

The x-terms cancel in R:

$$\begin{aligned} R &= X_1 X_3 - X_2^2 \\ &= \rho y^2 t^4. \end{aligned}$$

Therefore

$$\begin{aligned} R^3 z_4^2 &= \rho^3 d^2 y^{12} t^{18}, \\ z_1^6 &= \kappa^6 y^{12} t^{18}. \end{aligned}$$

Hence

$$R^3 z_4^2 = \lambda z_1^6, \qquad \lambda = \frac{\rho^3 d^2}{\kappa^6} = \frac{41 - 38i}{576} \neq 0. \tag{4}$$

This is the promised elimination certificate; it requires no division by y or t and therefore holds on the entire parameter space.

Proposition 8. *The critical-value closure is exactly the irreducible threefold*

$$V = V(F) \subset \mathbb{A}^4, \qquad F = (z_1 z_3 - z_2^2)^3 z_4^2 - \lambda z_1^6.$$

After a nonzero linear rescaling of z_4, one may equivalently take $\lambda = 1$.

Proof. Equation (4) shows that the image closure is contained in $V(F)$. It remains to prove irreducibility.

The polynomial

$$R = z_1 z_3 - z_2^2$$

is irreducible in $\mathbb{C}[z_1, z_2, z_3]$: it is primitive and linear in z_3, with relatively prime coefficient z_1 and constant term $-z_2^2$. Moreover $\gcd(R, z_1) = 1$, so the coefficients R^3 and $-z_1^6$ of $F = R^3 z_4^2 - z_1^6$ have no common nonunit factor. Thus F is primitive as a polynomial in z_4 over the UFD $\mathbb{C}[z_1, z_2, z_3]$. Over its fraction field, reducibility would require

$$\frac{\lambda z_1^6}{R^3}$$

to be a square. Its valuation along the irreducible prime R is -3, which is odd; hence it is not a square. Gauss' lemma gives irreducibility of F.

By Proposition 7, the image closure is irreducible of dimension three. The irreducible hypersurface $V(F)$ also has dimension three. An irreducible closed subset of an irreducible variety having the same dimension is the whole variety, so the critical-value closure equals $V(F)$. $\square$

4 The critical-value threefold is not a cylinder

Because nonzero target rescaling does not affect the argument, normalize $\lambda = 1$ and write

$$F = R^3 z_4^2 - z_1^6, \qquad R = z_1 z_3 - z_2^2.$$

Lemma 9. *The singular locus of V is*

$$\operatorname{Sing}(V) = P_1 \cup P_2,$$

where

$$P_1 = \{z_1 = z_2 = 0\}, \qquad P_2 = \{z_1 = z_4 = 0\}.$$

Their intersection is the line

$$L = P_1 \cap P_2 = \{z_1 = z_2 = z_4 = 0\}.$$

Proof. The partial derivatives are

$$\begin{aligned}
F_{z_4} &= 2R^3 z_4, \\
F_{z_3} &= 3R^2 z_1 z_4^2, \\
F_{z_2} &= -6R^2 z_2 z_4^2, \\
F_{z_1} &= 3R^2 z_3 z_4^2 - 6z_1^5.
\end{aligned}$$

At a singular point $F_{z_4} = 0$, hence either $R = 0$ or $z_4 = 0$.

If $R = 0$, the equation $F = 0$ gives $z_1 = 0$, and then $R = -z_2^2 = 0$, so $z_2 = 0$; this is P_1. Conversely every point of P_1 annihilates F and all first derivatives.

If $z_4 = 0$, the equation $F = 0$ gives $z_1 = 0$, yielding P_2, and again every point of P_2 is singular. Thus the singular locus is exactly $P_1 \cup P_2$. $\square$

Lemma 10. *For*

$$p_a = (0,0,a,0) \in L,$$

one has

$$\operatorname{mult}_{p_a} V = 5 \quad (a \neq 0), \qquad \operatorname{mult}_0 V = 6.$$

Thus the origin is the unique multiplicity-six point of L.

Proof. At p_a with $a \neq 0$, use local coordinates with $z_3 = a + u_3$. Then

$$R = az_1 + z_1 u_3 - z_2^2$$

has order one, with initial term az_1. Hence $R^3 z_4^2$ has order five and nonzero degree-five initial term $a^3 z_1^3 z_4^2$, whereas z_1^6 has order six. Therefore the local multiplicity is five.

At the origin, R has order two, so $R^3 z_4^2$ has order eight, while z_1^6 has order six. Therefore $\operatorname{mult}_0 V = 6$. □

Proposition 11. *The threefold V is not ambient-polynomially equivalent to a cylinder $Y \times \mathbb{A}^1 \subset \mathbb{A}^3 \times \mathbb{A}^1$.*

Proof. Assume that a polynomial automorphism Ψ of $\mathbb{A}^4$ sends V onto a cylinder $Y \times \mathbb{A}^1$. Translation in the $\mathbb{A}^1$ coordinate defines an algebraic $\mathbb{G}_a$-action on $\mathbb{A}^4$ preserving the cylinder and having no global fixed point. Conjugating by Ψ gives an algebraic $\mathbb{G}_a$-action on $\mathbb{A}^4$ preserving V and likewise having no global fixed point.

Every ambient polynomial automorphism Θ preserving V satisfies $\Theta(\operatorname{Sing}(V)) = \operatorname{Sing}(V)$, because regularity of the local ring is preserved under isomorphism. The planes P_1 and P_2 are the two irreducible components of $\operatorname{Sing}(V)$. An algebraic action of a connected algebraic group on a finite discrete set is trivial: each orbit is the connected image of the group and hence consists of one point. Since $\mathbb{G}_a$ is connected and the set of irreducible components $\{P_1, P_2\}$ is finite, each component is therefore preserved setwise. Therefore their intersection L is preserved setwise.

Local multiplicity is invariant under ambient polynomial automorphisms. By Lemma 10, the origin is the unique multiplicity-six point of L. Hence every element of the conjugated $\mathbb{G}_a$-action fixes the origin. This gives a global fixed point, contradicting the fixed-point-free translation action from which it was conjugated. Therefore no such ambient polynomial equivalence to a cylinder exists. □

5 Excluding all triangular one-coordinate suspensions

The determinant restrictions needed below are consequences, not assumptions. For a four-dimensional tangent sweep, Gao's Lemma 4.1 gives for the padded map

$$F_0(x, \gamma_T, a, b) = \bigl(\gamma_T x,\, T(\gamma_T, a, b)\bigr)$$

the identity

$$\det J(F_0) = \gamma_T^2 L + \gamma_T^3 M.$$

Here the exact padded/unpadded relation is immediate from the Jacobian: T is independent of x, so expansion along the x-column gives

$$\det J(F_0) = \gamma_T \, \det JT.$$

Consequently the corresponding unpadded sweep $T : \mathbb{A}^3 \to \mathbb{A}^3$ satisfies

$$\det JT = \gamma_T L + \gamma_T^2 M = \gamma_T (L + \gamma_T M), \tag{5}$$

where L, M are polynomials in the two non-γ_T base parameters.

Lemma 12 (Polynomial equivalence forces normalization)**.** *Let S be the sweep of Proposition 7, with source coordinate γ_S and*

$$\det JS = \gamma_S^2.$$

Let

$$\Sigma(u, v) = \big(T(u), Q(u, v)\big), \qquad u = (\gamma_T, a, b),$$

be any triangular one-coordinate suspension of a four-dimensional Gao sweep. If S and Σ are polynomially equivalent, then

$$\det J\Sigma = B\gamma_T^2$$

for some $B \in \mathbb{C}^\times$.

Proof. By symmetry of polynomial equivalence, write

$$S = \Psi \circ \Sigma \circ \Phi$$

for polynomial automorphisms Φ, Ψ of $\mathbb{A}^4$. Polynomial automorphisms have nonzero constant Jacobian determinant, so the chain rule and $\det JS = \gamma_S^2$ imply

$$\det J\Sigma = Cp^2, \qquad p := \gamma_S \circ \Phi^{-1}, \qquad C \in \mathbb{C}^\times.$$

The polynomial p is a coordinate polynomial, hence irreducible. Indeed the automorphism Φ^{-1} identifies $\mathbb{C}[p, p_2, p_3, p_4]$ with the full polynomial ring for suitable companion coordinates; equivalently, the quotient by (p) is isomorphic to a polynomial ring in three variables and is therefore a domain. Thus (p) is prime, in particular p is irreducible.

Because $\Sigma = (T, Q)$ is triangular and T is independent of v,

$$\det J\Sigma = (\partial_v Q) \det JT.$$

By (5), γ_T divides $\det JT$, hence

$$\gamma_T \mid \det J\Sigma = Cp^2.$$

The ring $\mathbb{C}[\gamma_T, a, b, v]$ is a UFD and the coordinate γ_T is irreducible, hence prime. Therefore $\gamma_T \mid p^2$ implies $\gamma_T \mid p$. Since p and γ_T are both irreducible, they are associates:

$$p = A\gamma_T, \qquad A \in \mathbb{C}^\times.$$

Consequently

$$\det J\Sigma = CA^2\gamma_T^2,$$

which is the asserted normalization. □

Lemma 13 (Triangular suspension exhaustiveness)**.** *Let*

$$\Sigma(u, v) = \big(T(u), Q(u, v)\big), \qquad u = (\gamma_T, a, b),$$

be a triangular one-coordinate suspension satisfying

$$\det J\Sigma = B\gamma_T^2, \qquad B \in \mathbb{C}^\times.$$

Write the general four-dimensional Gao sweep determinant as in (5)*. Then exactly one of the following two cases holds for some $A \in \mathbb{C}^\times$:*

(A) $M = 0$, $L = A$, hence

$$\det JT = A\gamma_T, \qquad Q = \frac{B}{A}\,\gamma_T v + h(u);$$

(B) $L = 0$, $M = A$, hence

$$\det JT = A\gamma_T^2, \qquad Q = \frac{B}{A}\,v + h(u).$$

In particular these two determinant modes are exhaustive; they are not part of the definition.

Proof. Because T is independent of v, the Jacobian matrix is block triangular:

$$J\Sigma = \begin{pmatrix} JT & 0 \\ \partial_u Q & \partial_v Q \end{pmatrix},$$

so

$$\det J\Sigma = (\partial_v Q) \det JT.$$

Using (5) and cancelling the nonzero factor γ_T in the integral domain $\mathbb{C}[\gamma_T, a, b, v]$ gives

$$(\partial_v Q)(L + \gamma_T M) = B\gamma_T. \tag{6}$$

The polynomial ring is a UFD and γ_T is irreducible. Hence the nonzero factor $L + \gamma_T M$ divides $B\gamma_T$. Since B is a unit, every nonunit irreducible factor of $L + \gamma_T M$ must be associated to γ_T. Moreover $L + \gamma_T M$ has degree at most one in γ_T. Therefore it is either a unit or an associate of γ_T.

If it is a unit, units in $\mathbb{C}[\gamma_T, a, b, v]$ are nonzero constants, so $M = 0$ and $L = A \in \mathbb{C}^\times$. Equation (6) gives $\partial_v Q = (B/A)\gamma_T$, and integration in v yields

$$Q = (B/A)\gamma_T v + h(u).$$

If $L + \gamma_T M$ is associated to γ_T, then $L = 0$ and $M = A \in \mathbb{C}^\times$. Equation (6) gives $\partial_v Q = B/A$, hence

$$Q = (B/A)v + h(u).$$

The two alternatives are mutually exclusive and exhaustive. $\square$

Lemma 14 (Critical-value invariant)**.** *If $S_2 = \Psi \circ S_1 \circ \Phi$ with polynomial automorphisms Φ, Ψ of $\mathbb{A}^4$, then*

$$\overline{\mathrm{CV}(S_2)} = \Psi\big(\overline{\mathrm{CV}(S_1)}\big).$$

In particular the dimension of the critical-value closure and the property of being ambient-polynomially equivalent to a cylinder are invariants of source-target polynomial equivalence.

Proof. Polynomial automorphisms have everywhere invertible Jacobian matrices. By the chain rule,

$$\mathrm{Crit}(S_2) = \Phi^{-1}(\mathrm{Crit}(S_1)).$$

Applying S_2 gives

$$\mathrm{CV}(S_2) = \Psi(\mathrm{CV}(S_1)),$$

and taking Zariski closures yields the formula. Dimension is preserved by Ψ, and cylinder-equivalence is preserved by composition with ambient polynomial automorphisms. $\square$

Proposition 15. *The sweep S of Proposition 7 is not polynomially equivalent to any triangular one-coordinate suspension of a four-dimensional Gao sweep.*

Proof. Suppose for contradiction that S is polynomially equivalent to a triangular one-coordinate suspension $\Sigma = (T, Q)$. Lemma 12 forces

$$\det J\Sigma = B\gamma_T^2, \qquad B \in \mathbb{C}^\times.$$

Lemma 13 then forces its determinant structure into exactly one of cases (A) or (B).

In case (A),

$$Q = \alpha\gamma_T v + h(u), \qquad \alpha \neq 0.$$

The set-theoretic critical locus is $\gamma_T = 0$. On that locus the v-dependence disappears from Q, so the critical-value set is parameterized by the two remaining base parameters. Hence

$$\dim \overline{\mathrm{CV}(\Sigma)} \leq 2.$$

Our sweep has a three-dimensional critical-value closure by Proposition 7. Lemma 14 excludes polynomial equivalence.

In case (B),

$$Q = \alpha v + h(u), \qquad \alpha \neq 0.$$

On the set-theoretic critical locus $\gamma_T = 0$, the coordinate v is free and the last target coordinate ranges over all of $\mathbb{A}^1$ for every base critical value. Thus

$$\overline{\mathrm{CV}(\Sigma)} = \overline{\mathrm{CV}(T)} \times \mathbb{A}^1,$$

a cylinder. Proposition 11 and Lemma 14 again exclude polynomial equivalence.

Lemmas 12 and 13 prove that any triangular one-coordinate suspension polynomially equivalent to S would be normalized automatically and would fall into exactly one of these two cases. Both cases are impossible. Hence no triangular one-coordinate suspension of a four-dimensional Gao sweep is polynomially equivalent to S. □

Proof of Theorem 3. Proposition 6 gives an exact polynomial middle-branch solution. Proposition 15 proves that its associated sweep is not polynomially equivalent to any triangular one-coordinate suspension of a four-dimensional Gao sweep. This is precisely the assertion. □

Proof of Corollary 4. Under the stated interpretation of Gao's phrase "equivalent to a suspension of a four-dimensional sweep," Theorem 3 supplies the polynomial solution whose existence is asked for in Open Problem 4.8. Hence the conjectured rigidity fails. □

6 Scope and verification

The main theorem is intentionally precise about the word "suspension." As emphasized in Remark 5, it excludes the formally defined triangular one-coordinate Gao-suspension class and does not assert a classification of every possible informal use of that word. Definition 2 imposes only the natural triangular one-coordinate form with a four-dimensional Gao base sweep; it imposes no determinant normalization. Lemma 12 proves that polynomial equivalence to our square-Jacobian sweep would itself force $\det J\Sigma \sim \gamma_T^2$. Lemma 13, using Gao's general four-dimensional sweep formula (Lemma 4.1), then proves that the two possibilities $\det JT \sim \gamma_T$ and $\det JT \sim \gamma_T^2$ are exhaustive. Gao's Theorem 4.6 supplies a concrete instance of case (A). Remark 4.7 motivates the graph-coordinate type of collapse but is not used as a formal classification theorem for case (B).

All witness identities in Propositions 6 and 7, the rank-three minor, and the implicit relation (4) have been checked in exact symbolic arithmetic. The accompanying verification script is intended as a reproducibility aid; the proof above does not depend on randomized computation.

References

[1] S. Gao, *Counterexamples to the Jacobian conjecture in dimensions greater than two*, arXiv:2608.00222v1 [math.AG], 31 July 2026. Relevant items: Lemma 4.1, equation (12), Theorem 4.6, Remark 4.7, Open Problem 4.8.